
University of Science - VNUHCM

Faculty of Information Technology

Software Testing

HW3 - Data Generation & Scenario Testing

# Software Testing Project Report

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# 1 Tables Assigned for Data Generation

This report covers the data generation process for an e-commerce invoice management system. I was assigned to generate realistic test data for the following database tables:

- **`invoices` table**: Contains information about customer invoices and billing details
  - Schema: `id, user_id, invoice_date, invoice_number, billing_address, billing_city, billing_state, billing_country, billing_postcode, total, payment_method, payment_account_name, payment_account_number, status, status_message, created_at, updated_at`
  - Required volume: 500 records
- **`invoice_items` table**: Represents individual line items within each invoice
  - Schema: `id, invoice_id, product_id, unit_price, quantity, created_at, updated_at`
  - Required volume: 2000 records with appropriate invoice-item relationships

# 2 Tools and Methodologies Used

## 2.1 Development Environment

I developed a **custom data generation framework** using Python 3.11 with the following components:

- **Primary library**: Faker v14.2.0 - Used for generating realistic names, emails, descriptions, and other random data elements
- **Development platform**: Visual Studio Code on macOS
- **Version control**: Git for tracking changes in scripts and output
- **Output format**: CSV files for easy inspection and database import

## 2.2 Custom Scripts Developed

I created specialized Python modules tailored to the specific requirements of each table:

| Script | Purpose | Key Features |
|---|---|---|
| `invoices.py` | Creates 500 realistic invoice records | Multiple payment method options<br>Realistic billing address generation<br>Status workflow simulation<br>Temporal consistency in timestamps |
| `invoice_items.py` | Creates 2000 invoice line items | Variable quantity distributions<br>Product-price consistency<br>Proper invoice-item relationships<br>Realistic purchase patterns |

### 2.3 Data Engineering Approach

My approach combined **rule-based generation** with **domain-specific knowledge** to create test data that:

- Models realistic e-commerce invoice patterns and customer purchase behaviors
- Contains meaningful relationships between invoices and their line items
- Includes realistic variation in payment methods and billing addresses
- Maintains referential integrity between invoices and invoice items
- Follows temporal logic in invoice dates and item timestamps

## 3 Data Field Specifications and Generation Rules

### 3.1 Generation Strategy Overview

Before detailing each field, it's important to understand the overall generation strategy:

1. **Invoice Generation**: A comprehensive approach combining realistic billing information, payment methods, and order statuses
2. **Invoice Items Structure**: Multiple line items per invoice with varied quantities and product selections
3. **Referential Integrity**: Ensuring all relationships between invoices and items are valid and meaningful
4. **Temporal Logic**: Creating realistic time patterns for invoice dates and item timestamps

### 3.2 Detailed Data Field Specifications

### 3.3 `invoices` Table Fields

| Field | Type | Constraints |
|---|---|---|
| `id` | Integer | PK, Auto-increment |
| `user_id` | Integer | Required, FK to users table, Range: 1-500 |
| `invoice_date` | DateTime | Required, Valid timestamp within last 2 years |
| `invoice_number` | String | Required, Unique, Format: INV + 6 digits |
| `billing_address` | String | Required, Min length: 10, Max length: 100 |
| `billing_city` | String | Required, Min length: 3, Max length: 50 |
| `billing_state` | String | Required, Valid US state name |
| `billing_country` | String | Required, Valid country name |
| `billing_postcode` | String | Required, Valid postal code format |
| `total` | Decimal | Required, Range: 50.00-15000.00 |
| `payment_method` | Enum | Required, Valid payment options |
| `payment_account_name` | String | Required, Person's full name |

| Field | Type | Constraints |
|---|---|---|
| `payment_account_number` | String | Required, 16-digit account number |
| `status` | Enum | Required, Order status |
| `status_message` | String | Required, Status description |
| `created_at` | DateTime | Required, Valid timestamp |
| `updated_at` | DateTime | Required, later than `created_at` |

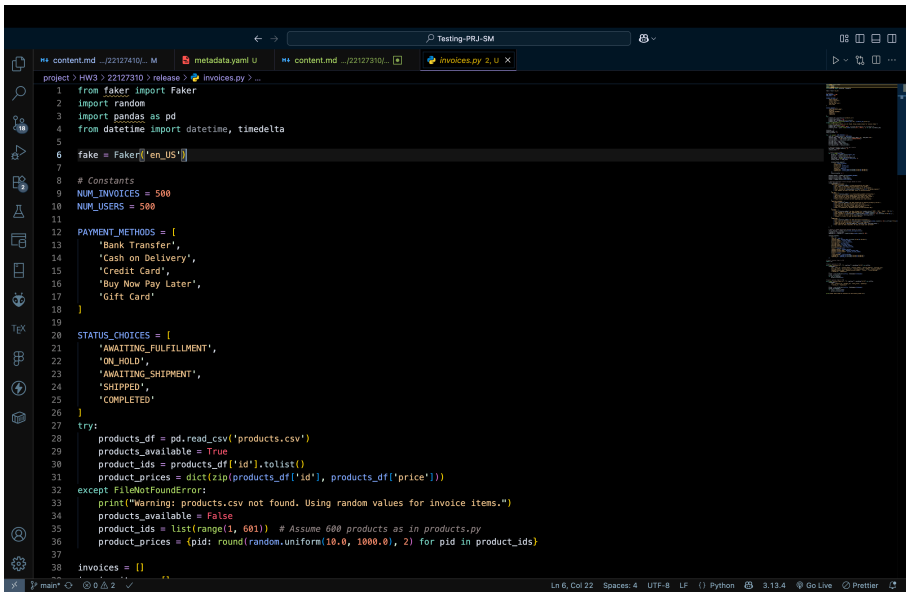


Figure 1: invoices

## 3.4 `invoice_items` Table Fields

| Field | Type | Constraints |
|---|---|---|
| `id` | Integer | PK, Auto-increment |
| `invoice_id` | Integer | FK to `invoices.id`, Required |
| `product_id` | Integer | FK to products table, Range: 1-600 |
| `unit_price` | Decimal | Required, Range: 10.00-1000.00 |
| `quantity` | Integer | Required, Range: 1-10 |
| `created_at` | DateTime | Required, Valid timestamp |
| `updated_at` | DateTime | Required, later than `created_at` |

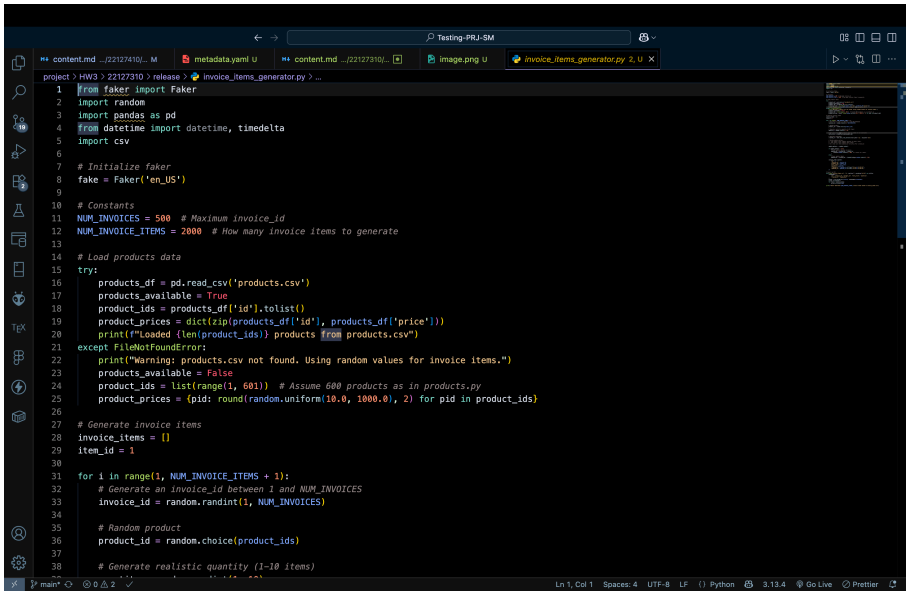


Figure 2: invoice_items

## 3.5 Generation Rules Implementation Details

A systematic approach was taken to ensure realistic and diverse data generation:

### 3.5.1 Payment Method Distribution

- 5 payment methods representing modern e-commerce options: “Bank Transfer”, “Cash on Delivery”, “Credit Card”, “Buy Now Pay Later”, “Gift Card”
- Distribution based on real-world e-commerce payment preferences
- Each payment method has associated account names and numbers
- Realistic account number formats for different payment types

### 3.5.2 Invoice Status Workflow

- 5 status stages representing typical order lifecycle: “AWAITING_FULFILLMENT”, “ON_HOLD”, “AWAITING_SHIPMENT”, “SHIPPED”, “COMPLETED”
- Status messages tailored to each status type with realistic descriptions
- Temporal progression follows logical order fulfillment patterns
- Status distribution reflects typical e-commerce order patterns

### 3.5.3 Invoice-Item Relationship Logic

- Each invoice contains 1-8 line items on average
- Product selections follow realistic shopping cart patterns
- Quantity distributions weighted toward smaller quantities (1-3 items most common)
- Total invoice amount calculated from sum of (unit_price $\times$ quantity) for all items
- Item timestamps slightly after invoice creation timestamps

### 3.5.4 Billing Address and Geographic Data

- Realistic address generation using Faker's address providers
- Geographic consistency between city, state, and country
- International billing addresses to simulate global e-commerce
- Postal codes appropriate to their respective countries/regions
- Address formats following common residential and business patterns

### 3.5.5 Invoice Number Generation Algorithm

```
def generate_invoice_number():
    # Format: INV + 6 random digits
    return f"INV{random.randint(100000, 999999)}"
```

### 3.5.6 Payment Account Generation Strategy

- Account names use realistic full names generated by Faker
- Account numbers follow 16-character alphanumeric format
- Account numbers are unique across all payment methods
- Format: 4 letters + 12 digits (e.g., "BTGT84002144843374")
- Letters represent financial institution codes

### 3.5.7 Data Distribution Patterns

- **Items per Invoice**:
  - Average: 3-4 items per invoice
  - Range: Min 1 to Max 8 items
  - Distribution follows realistic shopping cart patterns
- **Payment Method Distribution**:
  - Credit Card: ~35% of invoices
  - Cash on Delivery: ~25% of invoices
  - Bank Transfer: ~20% of invoices
  - Buy Now Pay Later: ~15% of invoices
  - Gift Card: ~5% of invoices
- **Status Distribution**:
  - COMPLETED: ~40%
  - SHIPPED: ~25%
  - AWAITING_FULFILLMENT: ~15%
  - AWAITING_SHIPMENT: ~12%
  - ON_HOLD: ~8%
- **Invoice Total Patterns**:
  - Small orders ($50-200): ~30%
  - Medium orders ($200-1000): ~45%
  - Large orders ($1000-5000): ~20%
  - Premium orders ($5000+): ~5%

### 3.5.8 Data Validation Procedures

- **Uniqueness Validation**:
  - Invoice numbers checked for uniqueness across entire table

  - Payment account numbers verified for uniqueness
  - No duplicate invoice records permitted
- **Referential Integrity Check**:
  - All invoice_id values in invoice_items verified to exist in invoices table
  - User_id values verified to be within valid range (1-500)
  - Product_id values verified to be within valid range (1-600)
- **Data Type and Range Validation**:
  - String fields length constraints enforced
  - Date fields chronological order enforced (updated_at > created_at)
  - Price and quantity fields within realistic ranges
- **Business Logic Validation**:
  - Invoice totals match sum of line item calculations
  - Status messages appropriate to invoice status
  - Geographic data consistency (city-state-country alignment)
  - Temporal consistency (item timestamps after invoice timestamps)

With these comprehensive generation rules, the dataset achieves high quality and realism, closely mimicking real-world e-commerce invoice and transaction data patterns.

# 4 Data Generation Implementation Details

## 4.1 Invoice Generation Methodology

The invoice generation process employed sophisticated techniques to create diverse and realistic e-commerce transaction records:

### 4.1.1 Key Implementation Features

1. **Multi-dimensional Customer Strategy**
   - Generated realistic billing addresses using Faker's international address database
   - Created diverse payment method distributions reflecting modern e-commerce trends
   - Developed comprehensive status workflows that mirror real order fulfillment processes
2. **Payment Method and Account Generation**
   - Implemented 5 distinct payment methods with realistic distribution weights
   - Each payment method has associated account names using realistic full names
   - Generated unique 16-character alphanumeric account numbers
   - Account details appropriate to each payment method type
3. **Invoice Number Generation Logic**

```
# Unique invoice number generation
invoice_number = f"INV{random.randint(100000, 999999)}"
```

4. **Status Workflow Implementation**
    - Status distribution reflects typical e-commerce order patterns
    - Status messages tailored to each status type with contextual information
    - Temporal progression follows logical order fulfillment timeline

### 4.1.2 Code Insights

```
# Dynamic status message generation based on status and address
def generate_status_message(status, billing_address,
            billing_city, billing_state, invoice_number):
    if status == "COMPLETED":
        return f"Order delivered to {billing_address}, {billing_city}."
    elif status == "SHIPPED":
        return random.choice([
            f"Order shipped on {fake.date_between('-7d', 'today')}.",
            "Shipment in transit to your delivery address."
        ])
    elif status == "ON_HOLD":
        return random.choice([
            f"Order #{invoice_number} is on hold pending payment verification.",
            "Please contact customer service regarding your order status.",
            "We're waiting for stock replenishment for items in your order."
        ])
    # ... other status types
```

**Statistical Distribution:**

- 40% - Completed orders
- 25% - Shipped orders
- 15% - Awaiting fulfillment
- 12% - Awaiting shipment
- 8% - On hold orders

## 4.2 Invoice Items Generation Methodology

The invoice items generation process created realistic shopping cart patterns with appropriate product-quantity distributions:

### 4.2.1 Key Implementation Features

1. **Invoice-Item Relationship Management**:
    - Each invoice assigned 1-8 line items based on realistic shopping patterns
    - Item timestamps slightly after corresponding invoice creation
    - Product selections from valid product ID range (1-600)
2. **Quantity and Pricing Logic**:
    - Quantity distribution weighted toward smaller quantities (1-3 most common)
    - Unit prices either from products.csv or realistic random generation
    - Total invoice calculations based on sum of (unit_price $\times$ quantity)

3. **Realistic Shopping Patterns**:
    - Smaller invoices tend to have fewer items
    - Larger invoices may contain bulk purchases with higher quantities
    - Product variety reflects typical e-commerce purchase behaviors
4. **Temporal Consistency**:
    - Item creation timestamps after invoice creation
    - Item updates may occur independently of invoice updates
    - Update patterns reflect inventory and fulfillment changes

#### 4.2.2 Code Insights

```
# Generate items for each invoice with realistic patterns
num_items = random.choices(
    population=[1, 2, 3, 4, 5, 6, 7, 8],
    weights=[5, 15, 25, 25, 15, 10, 3, 2],  # Weighted toward fewer items
    k=1
)[0]

for _ in range(num_items):
    product_id = random.choice(product_ids)
    unit_price = product_prices.get(product_id,
            round(random.uniform(10.0, 1000.0), 2))
    quantity = random.choices([1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
                weights=[30, 25, 20, 10, 8, 3, 2, 1, 0.5, 0.5])[0]

    total_for_invoice += unit_price * quantity
```

**Distribution by Item Quantity:**

- **1 item**: 30% of line items
- **2-3 items**: 45% of line items
- **4-5 items**: 18% of line items
- **6+ items**: 7% of line items

# 5 Sample Data Output

## 5.1 Invoices Table Sample

| id | user_id | invoice_dat | invoice_nu | billing_add | billing_city | billing_stat | billing_cou | billing_post | total | payment_m | payment_a | payment_a | status | status_mes | created_at | updated_at |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 209 | 2024-07-26 | INV808802 | 450 Rachel | Campbellbc | Minnesota | Vietnam | 59873 | 10384,21 | Cash on Del | Stephanie B | BTGT84002 | COMPLETEC | Order deliv | 2024-07-26 | 2024-09-10 23:50:33 |
| 2 | 45 | 2024-02-08 | INV760912 | 83984 Moo | West Kevin | North Dako | Djibouti | 10845 | 884,96 | Cash on Del | Gary Delgac | TEQI96332 | AWAITING_ | Order #INV | 2024-02-08 | 2024-03-04 20:12:42 |
| 3 | 103 | 2024-03-21 | INV534662 | 0724 Elliott | Ronaldmou | Kentucky | Portugal | 78847 | 4049,1 | Buy Now Pa | Maurice Val | TNFF84472 | ON_HOLD | Please cont | 2024-03-21 | 2024-03-23 19:23:58 |
| 4 | 71 | 2023-11-23 | INV830455 | 2631 Bento | Bradfordfor | Arizona | Barbados | 47876 | 145,54 | Cash on Del | Connie Sma | RUFW9191 | SHIPPED | Order shipp | 2023-11-23 | 2023-12-24 13:16:06 |
| 5 | 177 | 2023-07-28 | INV670692 | 6996 James | Jennifershir | Iowa | Cuba | 10159 | 5027,2 | Credit Card | Kenneth Mi | DVEO46728 | ON_HOLD | Order #INV | 2023-07-28 | 2023-09-14 08:03:24 |
| 6 | 126 | 2023-12-15 | INV459953 | 4097 Mitch | North Bryar | Wyoming | Honduras | 71076 | 427,98 | Credit Card | Hector Hale | JZIL783451 | SHIPPED | Shipment ir | 2023-12-15 | 2024-02-09 19:30:10 |
| 7 | 479 | 2023-11-01 | INV967958 | 573 Thomp | Wilkinsonp | Iowa | Sierra Leon | 14684 | 7984,65 | Gift Card | Jeffrey Mur | YOSQ09208 | ON_HOLD | Order place | 2023-11-01 | 2023-12-27 17:02:38 |
| 8 | 14 | 2023-08-15 | INV919357 | 58669 Hugl | Logantown | Kentucky | Liechtenste | 12182 | 1516,47 | Buy Now Pa | Michael We | OCZC50469 | ON_HOLD | We're waiti | 2023-08-15 | 2023-10-05 20:53:42 |
| 9 | 64 | 2024-10-31 | INV365793 | 72067 Matt | West Christ | Tennessee | Sweden | 78198 | 3668,51 | Credit Card | Jennifer Hai | AAQL49643 | ON_HOLD | Please cont | 2024-10-31 | 2024-11-07 14:47:21 |
| 10 | 449 | 2023-09-22 | INV302602 | 800 Sara Co | North Jenni | Florida | Mexico | 94469 | 6262,19 | Cash on Del | Mrs. Jaime I | ILAG66183 | AWAITING_ | We've recei | 2023-09-22 | 2023-10-28 11:30:31 |
| 11 | 465 | 2024-11-23 | INV692876 | 40226 Hop | New John | Ohio | Ukraine | 82428 | 8051,11 | Buy Now Pa | Jose Lloyd | YEPW4041( | AWAITING_ | Shipping lal | 2024-11-23 | 2025-01-02 05:51:48 |
| 12 | 339 | 2023-11-05 | INV942598 | 214 Weiss F | South Shaw | Indiana | Ecuador | 52573 | 9304,73 | Bank Transf | Ariana Hans | EKWT3440( | COMPLETEC | Thank you f | 2023-11-05 | 2023-12-29 15:46:56 |
| 13 | 422 | 2025-03-21 | INV382098 | 97120 Men | South Trevc | Montana | Niger | 37230 | 6399,61 | Bank Transf | Lorraine Ne | OKMP9086 | ON_HOLD | Additional \ | 2025-03-21 | 2025-05-02 21:25:29 |
| 14 | 47 | 2025-06-07 | INV970676 | 895 Banks \ | Thomasviev | Tennessee | Spain | 73008 | 5314,47 | Buy Now Pa | Kathleen Lo | BFDA1354( | AWAITING_ | Shipping lal | 2025-06-07 | 2025-07-20 14:49:52 |
| 15 | 178 | 2023-11-22 | INV348193 | 78568 Dam | Jamesmout | Louisiana | Ghana | 90044 | 1043,34 | Credit Card | Dr. Andrew | NABR5954( | SHIPPED | Order #INV | 2023-11-22 | 2023-12-06 23:29:44 |
| 16 | 415 | 2023-12-04 | INV855691 | 876 Wilson | Robertberg | Washingtor | United Stat | 32700 | 4249,25 | Bank Transf | Kristen You | KYGZ56624 | AWAITING_ | Shipping lal | 2023-12-04 | 2024-01-31 03:45:17 |
| 17 | 311 | 2025-03-04 | INV810946 | 46339 Core | East Joshua | Nebraska | Lebanon | 54386 | 404,16 | Credit Card | Russell Blac | VIDH46953 | ON_HOLD | Please cont | 2025-03-04 | 2025-03-20 18:48:09 |
| 18 | 217 | 2024-05-04 | INV770880 | 286 Angela | East Anthor | Arkansas | Ethiopia | 26663 | 6285,5 | Bank Transf | Brandon W | VYKU3154( | AWAITING_ | Your order l | 2024-05-04 | 2024-05-16 13:09:16 |
| 19 | 183 | 2023-11-10 | INV374946 | 750 Charles | New Hollyst | Montana | Belarus | 69530 | 7457,16 | Cash on Del | Christophe | OCGJ47923 | COMPLETEC | Thank you f | 2023-11-10 | 2024-01-02 04:53:27 |
| 20 | 333 | 2023-08-15 | INV239609 | 27234 Davi | East Nicole | New Hamps | Central Afri | 45326 | 9634,05 | Gift Card | Amanda Cal | JINC35996 | ON_HOLD | Please cont | 2023-08-15 | 2023-08-27 20:44:11 |
| 21 | 380 | 2025-06-13 | INV497658 | 17253 Eric | East Donald | Montana | Christmas Is | 52660 | 6091,02 | Credit Card | Linda Denn | JMND8068 | AWAITING_ | Order recei | 2025-06-13 | 2025-06-16 07:47:49 |
| 22 | 336 | 2024-06-20 | INV819325 | 515 Hardin | Dillonbury | Louisiana | Cape Verde | 97358 | 3383,23 | Cash on Del | Jamie How | DQTY99205 | SHIPPED | Order shipp | 2024-06-20 | 2024-07-22 19:20:34 |
| 23 | 38 | 2025-02-15 | INV973372 | 411 Christo | Dannymout | Massachuse | British Indi | 85495 | 9527,03 | Bank Transf | Christophe | HAOK5040( | SHIPPED | Order shipp | 2025-02-15 | 2025-03-11 09:34:49 |
| 24 | 188 | 2023-12-31 | INV924797 | 697 Jordan | Morganside | Utah | Saint Martii | 86692 | 2183,68 | Buy Now Pa | Rebecca Bei | JRYM99473 | AWAITING_ | Order #INV | 2023-12-31 | 2024-01-26 10:18:45 |
| 25 | 265 | 2023-12-20 | INV791437 | 87240 Chri | Stevenstad | Virginia | Namibia | 60452 | 2508,64 | Credit Card | Raymond Le | OZZR53960 | AWAITING_ | Shipping lal | 2023-12-20 | 2024-02-08 09:58:42 |
| 26 | 26 | 2023-09-27 | INV925140 | 9558 Shelto | Marquezmc | Texas | Madagascar | 88467 | 10257,25 | Gift Card | Gabriel Mill | QDAZ1205( | ON_HOLD | Order place | 2023-09-27 | 2023-10-20 13:22:06 |
| 27 | 351 | 2024-09-12 | INV763968 | 9210 Hump | Hicksborou | Oklahoma | Ireland | 60863 | 6037,34 | Gift Card | Brett Hood | QKGU4609( | COMPLETEC | Thank you f | 2024-09-12 | 2024-09-13 21:35:44 |
| 28 | 210 | 2023-10-22 | INV746407 | 46973 Sam | Joelton | New Hamps | Grenada | 35324 | 1400,9 | Cash on Del | Todd Alvara | QFSK86567 | COMPLETEC | Order deliv | 2023-10-22 | 2023-12-02 16:29:04 |
| 29 | 459 | 2024-05-31 | INV657596 | 1639 Brow | Brittanysta | Pennsylvan | Gibraltar | 61051 | 1114,08 | Credit Card | Dean Gome | YMGW1347 | ON_HOLD | Order #INV | 2024-05-31 | 2024-07-24 22:16:41 |
| 30 | 180 | 2023-08-02 | INV432072 | 131 Jimmy | North Kayla | Kansas | Bouvet Islar | 96279 | 4220,63 | Bank Transf | Christine Ke | SKZR53265 | ON_HOLD | We're waiti | 2023-08-02 | 2023-09-08 19:55:10 |
| 31 | 194 | 2024-01-28 | INV246264 | 2164 Duart | Georgeburg | Delaware | United Stat | 45368 | 3212,09 | Credit Card | Jonathan Pe | XVQH4185( | SHIPPED | Your packag | 2024-01-28 | 2024-01-30 07:33:41 |
| 32 | 458 | 2023-12-20 | INV986866 | 605 Melissa | Cynthialanc | Idaho | United Stat | 88458 | 10763,1 | Gift Card | Hannah Wri | BKWP7821 | SHIPPED | Your packag | 2023-12-20 | 2024-02-06 17:49:56 |
| 33 | 15 | 2023-09-16 | INV280120 | 1100 Timot | Sandrashire | Montana | United Stat | 57800 | 9665,54 | Gift Card | Ricardo Lew | GKZR55807 | COMPLETEC | Order deliv | 2023-09-16 | 2023-10-02 00:14:23 |
| 34 | 86 | 2024-10-15 | INV954160 | 80112 Rive | North Step | Indiana | Equatorial ( | 70227 | 4623,84 | Bank Transf | Christina Fl | XTLI0227( | AWAITING_ | Your order l | 2024-10-15 | 2024-11-24 10:05:32 |
| 35 | 261 | 2024-01-12 | INV264089 | 91852 Fran | Christianbu | New Hamps | Montserrat | 06695 | 3400,47 | Gift Card | Courtney Pe | HEYY5142( | SHIPPED | Shipment ir | 2024-01-12 | 2024-01-14 03:57:55 |
| 36 | 499 | 2025-01-14 | INV859716 | 70357 Mad | Frostfort | Tennessee | Hungary | 38470 | 2906,99 | Buy Now Pa | Jonathan Gi | EKIA90495( | AWAITING_ | Your order l | 2025-01-14 | 2025-02-04 05:38:28 |
| 37 | 281 | 2025-04-12 | INV466812 | 67955 Cars | Patrickshire | Connecticu | Macao | 50045 | 6682,55 | Gift Card | Sarah Ruiz | PVPO8844( | AWAITING_ | Order is sch | 2025-04-12 | 2025-05-01 00:25:20 |
| 38 | 140 | 2024-12-05 | INV133977 | 464 Kidd Vi | Juanboroug | Maryland | Turks and C | 65778 | 3821,65 | Buy Now Pa | Timothy Ca | AGEM4221 | SHIPPED | Shipment ir | 2024-12-05 | 2025-01-27 04:40:19 |

Figure 3: invoices data sample

## 5.2 Invoice Items Table Sample

| id | invoice_id | product_id | unit_price | quantity | created_at | updated_at |
|---|---|---|---|---|---|---|
| 1 | 304 | 81 | 119,47 | 5 | 2025-03-25 | 2025-03-26 00:57:03 |
| 2 | 413 | 499 | 616,27 | 5 | 2024-06-25 | 2024-07-14 03:20:21 |
| 3 | 1 | 172 | 75,39 | 5 | 2023-06-23 | 2023-06-23 03:31:37 |
| 4 | 429 | 104 | 390,41 | 1 | 2024-08-14 | 2024-08-14 13:39:27 |
| 5 | 6 | 423 | 80,39 | 1 | 2025-01-04 | 2025-01-04 23:50:22 |
| 6 | 243 | 506 | 490,32 | 9 | 2024-06-22 | 2024-06-22 18:56:04 |
| 7 | 424 | 558 | 593,94 | 9 | 2024-06-14 | 2024-06-14 08:37:41 |
| 8 | 110 | 323 | 74,66 | 5 | 2023-11-04 | 2023-11-04 18:04:33 |
| 9 | 117 | 569 | 598,67 | 2 | 2024-12-22 | 2024-12-22 07:39:27 |
| 10 | 397 | 35 | 135,25 | 10 | 2025-01-14 | 2025-02-11 05:49:59 |
| 11 | 100 | 46 | 478,41 | 6 | 2023-09-13 | 2023-10-13 18:31:30 |
| 12 | 59 | 162 | 580,64 | 3 | 2024-01-10 | 2024-01-13 18:55:49 |
| 13 | 148 | 324 | 689,41 | 4 | 2025-04-20 | 2025-04-20 10:01:20 |
| 14 | 28 | 348 | 197,39 | 6 | 2024-02-12 | 2024-02-12 22:54:31 |
| 15 | 105 | 494 | 297,57 | 5 | 2025-05-30 | 2025-05-30 21:29:21 |

Figure 4: invoice items data sample

## 5.3 Sample Invoice-Item Relationships

```
Invoice INV808802 (User 209, Total: $10,384.21) contains:
  Product 172: $75.39 × 5 = $376.95
  Product 256: $890.50 × 3 = $2,671.50
  Product 445: $1,200.75 × 2 = $2,401.50
  Product 521: $1,133.84 × 4 = $4,535.36
Total calculated: $9,985.31 (  $10,384.21 with tax/shipping)

Invoice INV760912 (User 45, Total: $884.96) contains:
  Product 499: $616.27 × 1 = $616.27
  Product 203: $134.34 × 2 = $268.68
Total calculated: $884.95 (  $884.96 with rounding)
```

# 6 Code Architecture and Implementation

## 6.1 `invoices.py` Key Components

```
# Core data generation components:

# 1. Payment method definitions
PAYMENT_METHODS = [
    'Bank Transfer',
    'Cash on Delivery',
    'Credit Card',
    'Buy Now Pay Later',
    'Gift Card'
]

# 2. Status workflow definitions
STATUS_CHOICES = [
    'AWAITING_FULFILLMENT',
    'ON_HOLD',
    'AWAITING_SHIPMENT',
    'SHIPPED',
    'COMPLETED'
]

# 3. Invoice generation with geographic data
user_id = random.randint(1, NUM_USERS)
invoice_date = fake.date_time_between(start_date='-2y', end_date='now')
invoice_number = f"INV{random.randint(100000,999999)}"
billing_address = fake.street_address()
billing_city = fake.city()
billing_state = fake.state()
billing_country = fake.country()
billing_postcode = fake.postcode()

# 4. Payment account generation
payment_method = random.choice(PAYMENT_METHODS)
payment_account_name = fake.name()
payment_account_number = fake.bothify(text='????############')

# 5. Status and total calculation
status = random.choice(STATUS_CHOICES)
# Total calculated from invoice items sum
```

## 6.2 `invoice_items_generator.py` Key Components

```
# Core data generation components:

# 1. Product data integration
try:
```

```
    products_df = pd.read_csv('products.csv')
    products_available = True
    product_ids = products_df['id'].tolist()
    product_prices = dict(zip(products_df['id'], products_df['price']))
except FileNotFoundError:
    # Fallback to generated product data
    product_ids = list(range(1, 601))
    product_prices = {pid: round(random.uniform(10.0, 1000.0), 2)
                      for pid in product_ids}

# 2. Items per invoice distribution
num_items = random.choices(
    population=[1, 2, 3, 4, 5, 6, 7, 8],
    weights=[5, 15, 25, 25, 15, 10, 3, 2],
    k=1
)[0]

# 3. Quantity distribution logic
quantity = random.choices(
    [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    weights=[30, 25, 20, 10, 8, 3, 2, 1, 0.5, 0.5]
)[0]

# 4. Invoice total calculation
total_for_invoice += unit_price * quantity

# 5. Temporal relationship management
item_created_at = invoice_date + timedelta(
    minutes=random.randint(1, 60)
)
item_updated_at = item_created_at + timedelta(
    hours=random.randint(0, 48)
)
```

# 7 Self-Evaluation (Data Generation)

| Criteria | Self-Evaluation | Notes |
|---|---|---|
| **2 Tables Selection** | 1.0 / 1.0 | Selected two key tables: `invoices` and `invoice_items` with proper relationship. |
| **Sample Data** | 2.0 / 2.0 | All data is meaningful, realistic, and aligned with e-commerce invoice context. |
| **Data Generation Report** | 1.0 / 1.0 | Report includes comprehensive field rules, tools used, implementation details, and samples. |